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Proof-of-principle demonstration of modified “Ping-Pong” protocol on telecom fiber

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Supervisor Zheng-fu Han

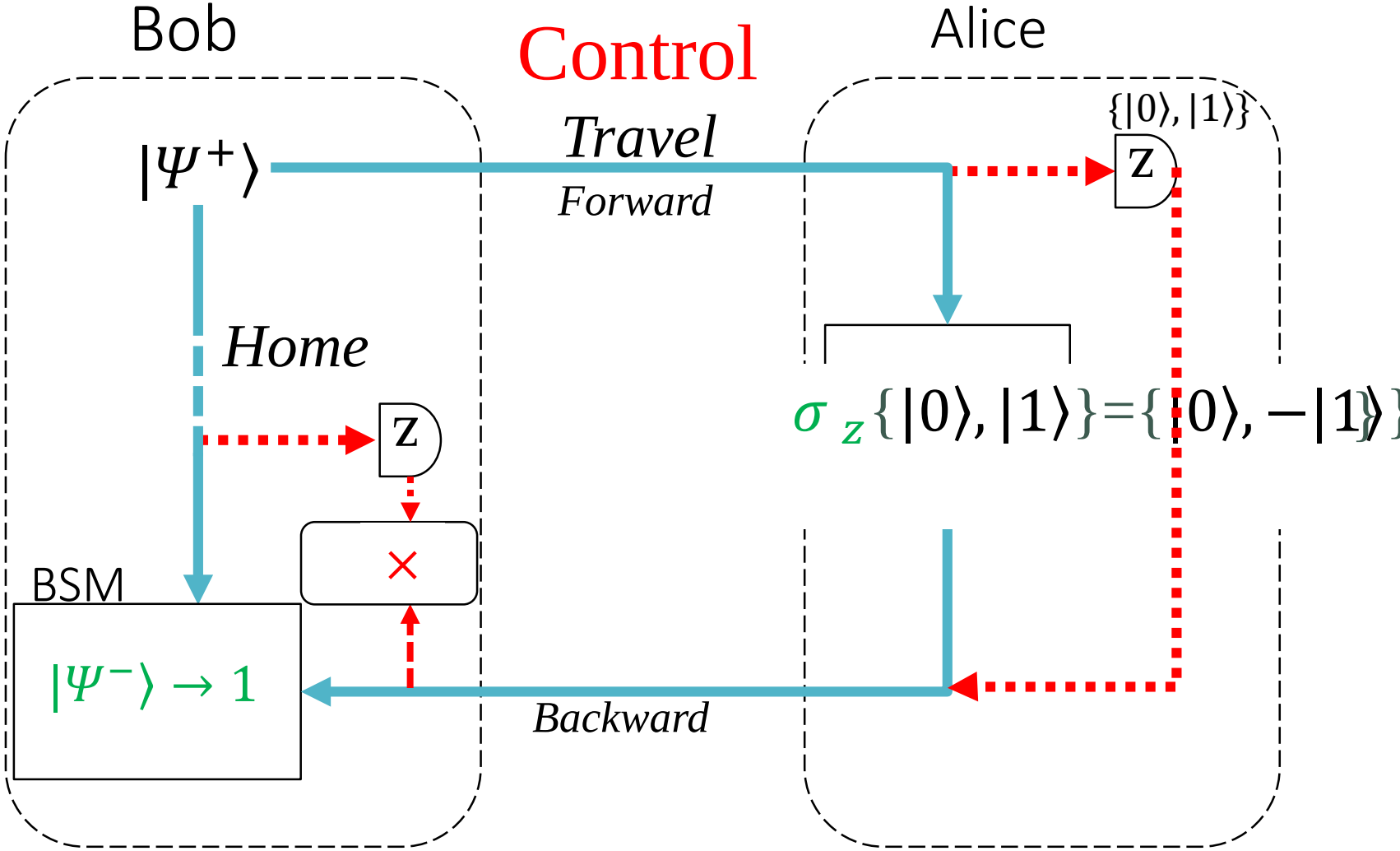
Content

Modified Ping-Pong protocol

Experiment setup

Experiment results

Ping-Pong protocol



Boström, K. & Felbinger, T. Deterministic Secure Direct Communication Using Entanglement. Phys. Rev. Lett. 89, 187902 (2002).

Security in lossy channel

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1. Wójcik, A. Eavesdropping on the ‘ping-pong’ quantum communication protocol. *Phys. Rev. Lett.* **90,** 157901 (2003).
2. Cai, Q. The ‘ping-pong’ protocol can be attacked without eavesdropping. *Phys. Rev. Lett.* **91,** 109801 (2003).
3. Zhang, Z., Man, Z. & Li, Y. Improving Wójcik’s eavesdropping attack on the ping-pong protocol. *Phys. Lett. Sect. A Gen. At. Solid State Phys.* **333,** 46–50 (2004).
4. Boström, K. & Felbinger, T. On the security of the ping-pong protocol. *Phys. Lett. Sect. A Gen. At. Solid State Phys.* **372,** 3953–3956 (2008).

Ping-Pong protocol

Major modification

I → $I_0\{|v\rangle, |0\rangle, |1\rangle\}=\{|v\rangle, |0\rangle, |1\rangle\}$

$\nexists$

$I_1\{|v\rangle, |0\rangle, |1\rangle\}=\{|v\rangle, -|0\rangle, -|1\rangle\}$

$|v\rangle$, the vacuum state

σ_z → $Y_0\{|v\rangle, |0\rangle, |1\rangle\}=\{|v\rangle, |0\rangle, -|1\rangle\}$

$\nexists$

$Y_1\{|v\rangle, |0\rangle, |1\rangle\}=\{|v\rangle, -|0\rangle, |1\rangle\}$

Han, Y.-G. et al. Security of modified Ping-Pong protocol in noisy and lossy channel. Sci. Rep. 4, 1–4 (2014).

Modified Ping-PONG

Secure key rate R :

$$R \geq 1 - \frac{S\left(\frac{p_{00}}{p_{00}+p_{10}}\right) + S\left(\frac{p_{11}}{p_{11}+p_{01}}\right)}{2\eta} - S(e),$$

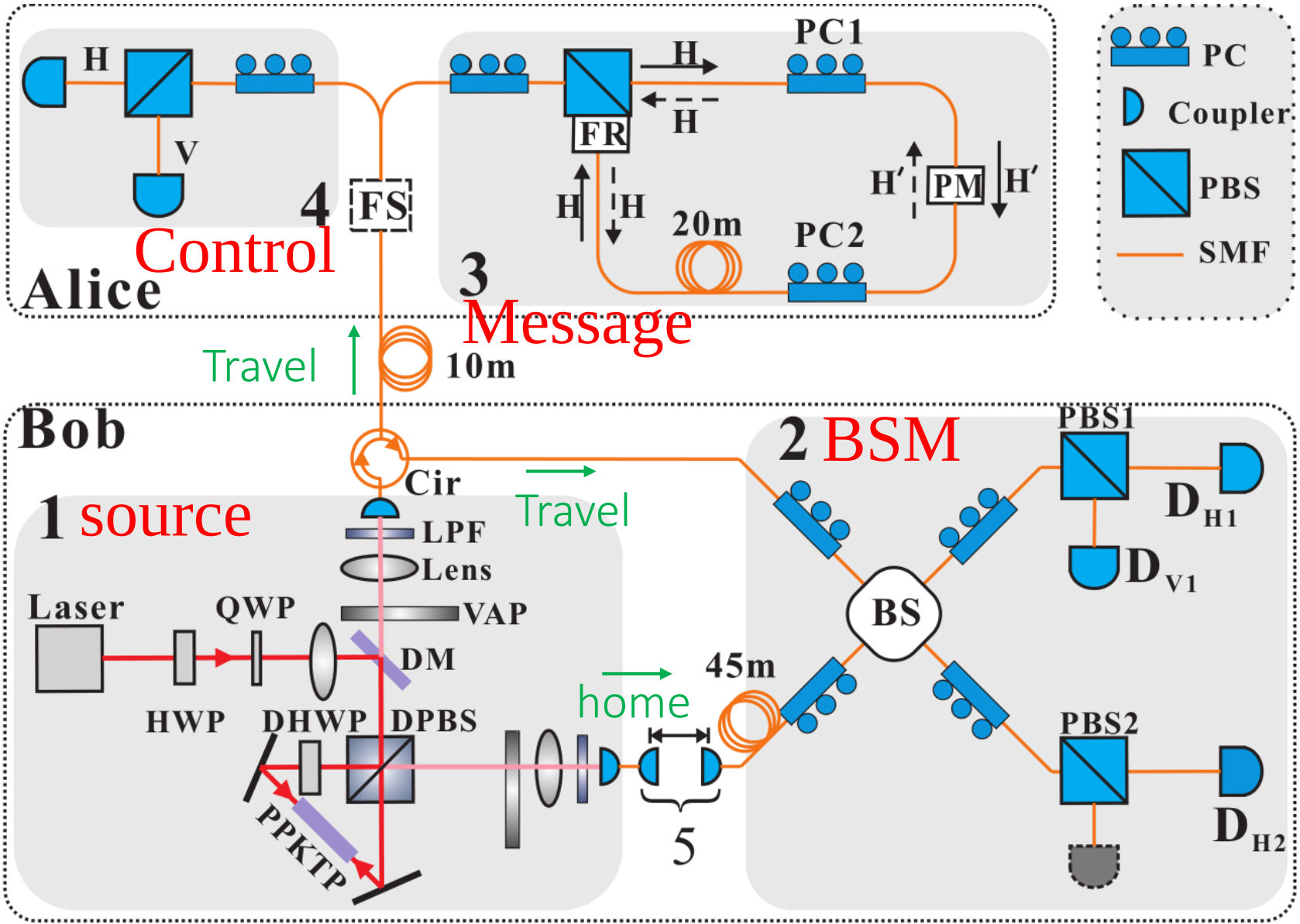
S : Shannon's binary entropy function.

p_{01} : the probability that Alice measures $|0\rangle$ when Bob measures $|1\rangle$

η: the transmission efficiency of the **backward** public channel

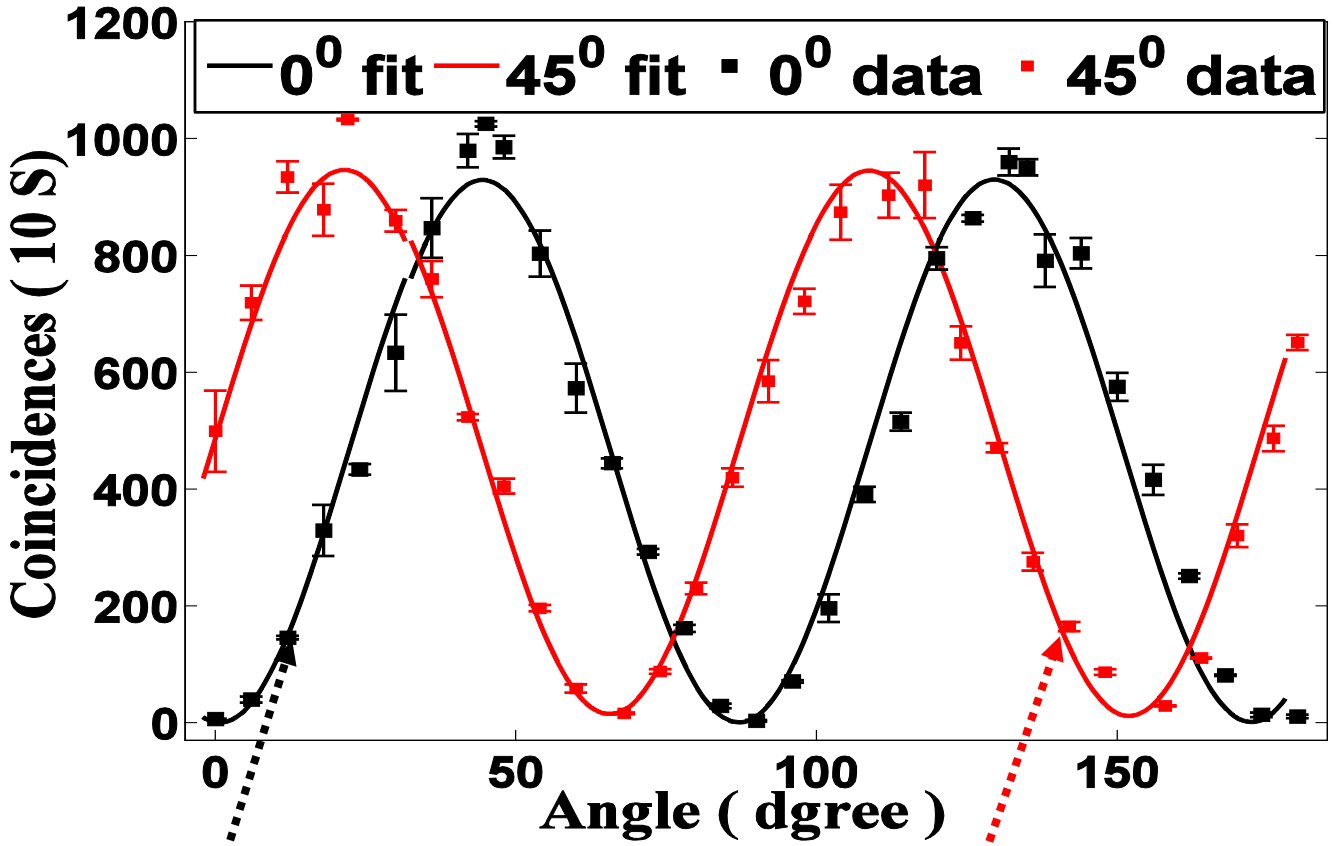
e: the error rate under the message mode

Experiment


H
V
4
FS
Control
Alice
PC1
H
H
FR
H
H
20m
PC2
3
H′
PM
H′
Message
Travel
10m
PC
Coupler
PBS
SMF
Bob
1 source
Cir
Travel
LPF
Lens
VAP
Laser
QWP
DM
HWP
DHWP
DPBS
PPKTP
home
45m
5
2 BSM
PBS1
D H1
D V1
BS
PBS2
D H2

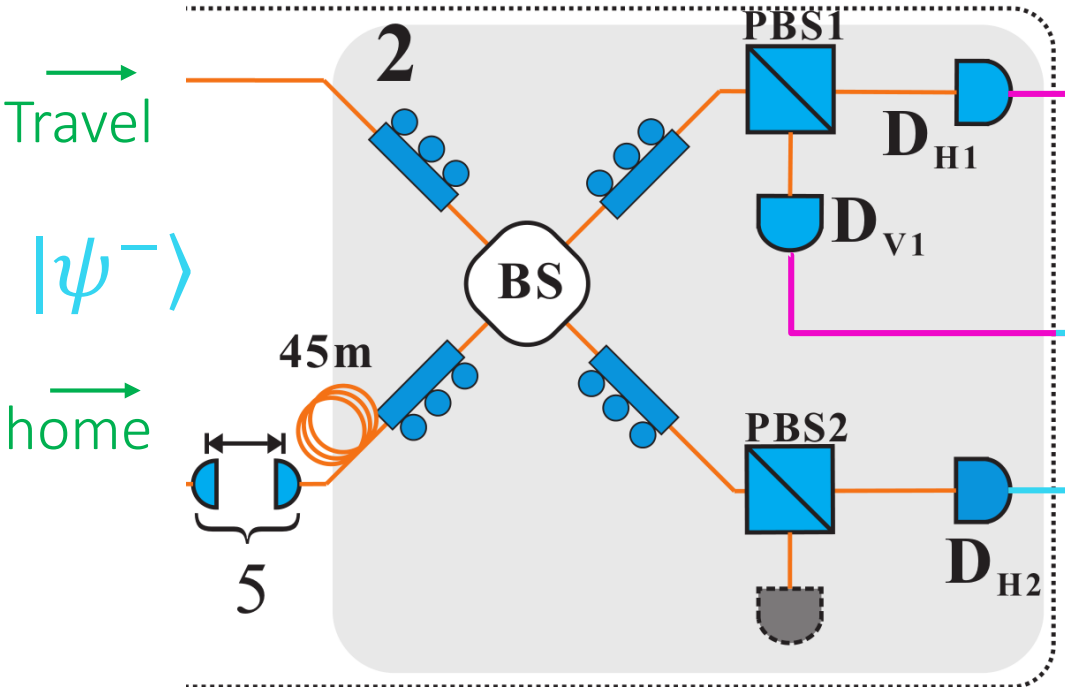
EPR photon source

Type II PPKTP + Sagnac interferometer + 1560-nm degenerate



Kim, Taehyun, Marco Fiorentino, and Franco NC Wong. "Phase-stable source of polarization-entangled photons using a polarization Sagnac interferometer." *Physical Review A* 73.1 (2006): 012316.

BSM



C_{V1H1}

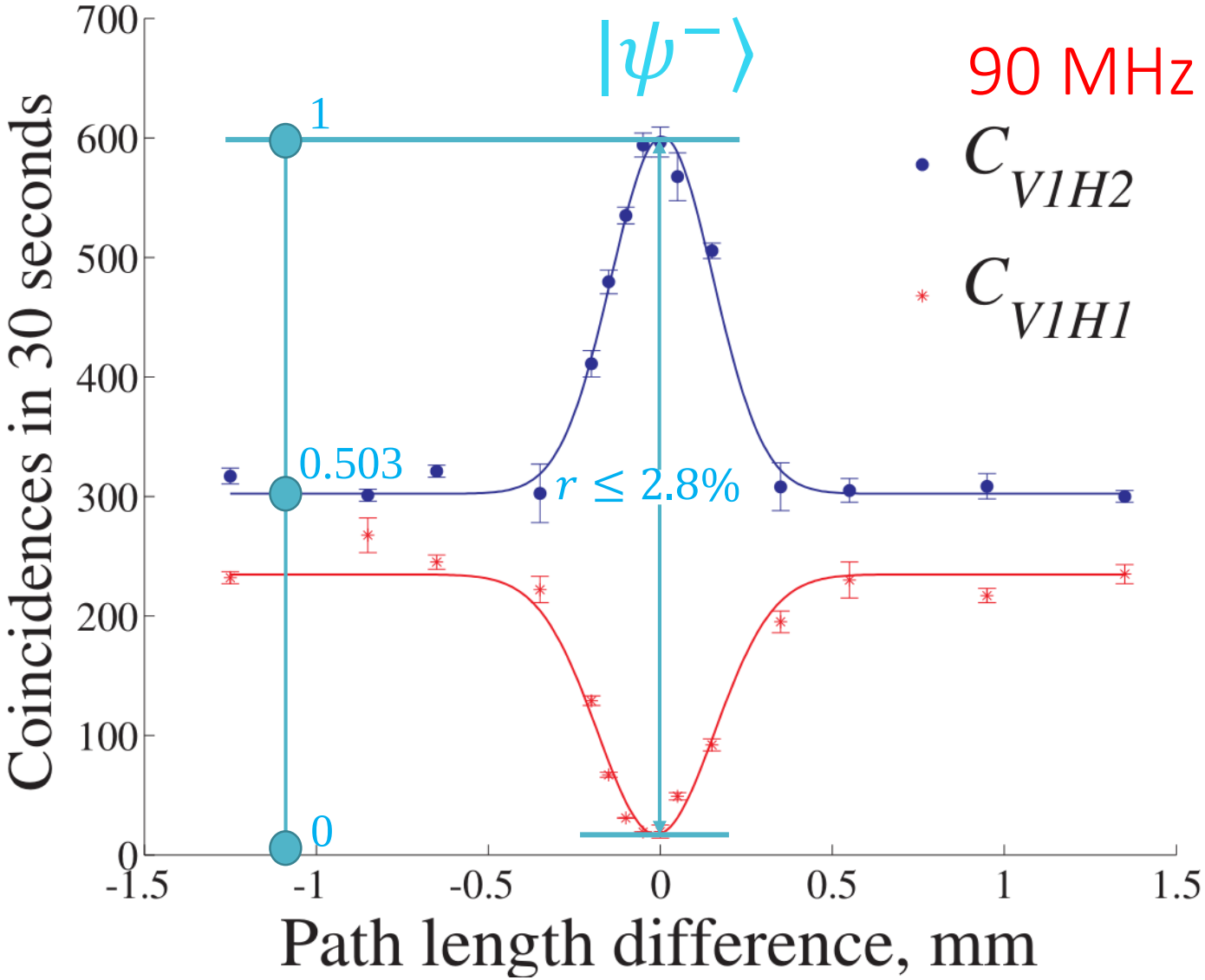
$\dfrac{}{} = r$

C_{V1H2}

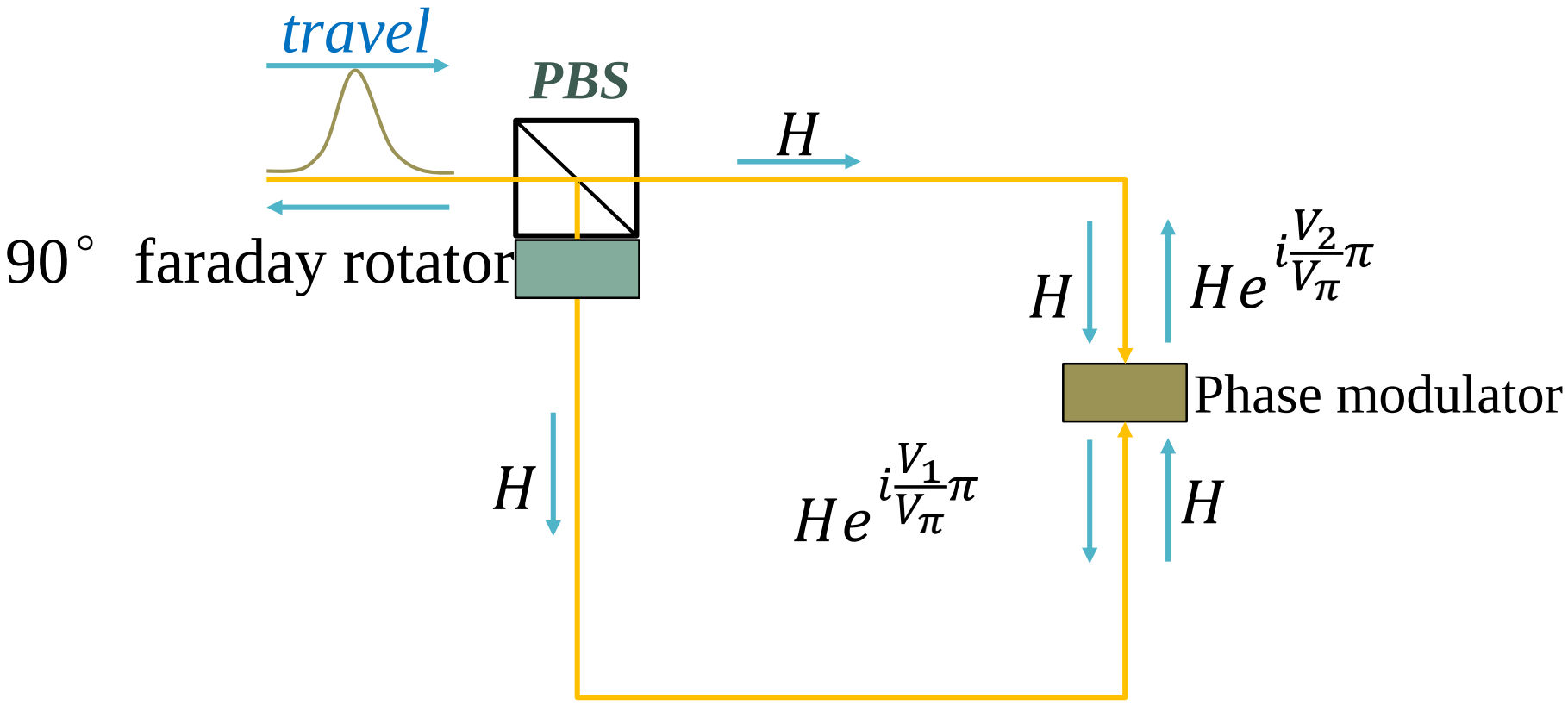
$|\psi^+\rangle \rightarrow r = \infty.$

$|\psi^-\rangle \rightarrow r = 0.$

HOM


$|\psi^-\rangle$
90 MHz
C_{V1H2}
C_{V1H1}
1
0.503
0
$r \leq 2.8\%$
Coincidences in 30 seconds
Path length difference, mm
700
600
500
400
300
200
100
0
-1.5
-1
-0.5
0
0.5
1
1.5

MESSAGE MODE: CODING



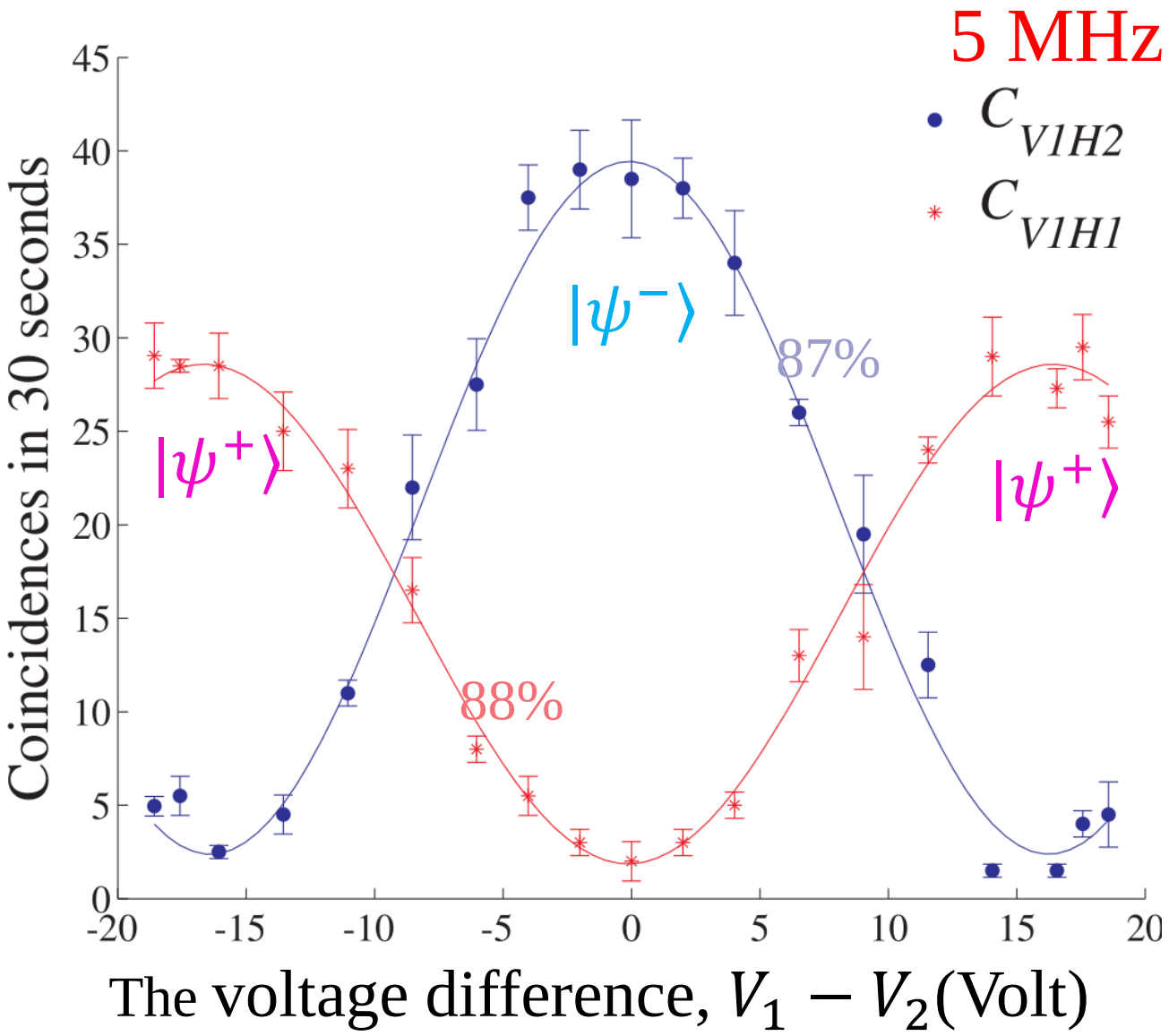
Operation: $F\{|v\rangle, |H\rangle, |V\rangle\} = \{|v\rangle, e^{i\frac{V_1}{V_\pi}\pi}|H\rangle, e^{i\frac{V_2}{V_\pi}\pi}|V\rangle\}$

BSM: $|\psi^-\rangle \rightarrow |H\rangle_h|V\rangle_t - e^{i\frac{V_1-V_2}{V_\pi}\pi}|V\rangle_h|H\rangle_t$

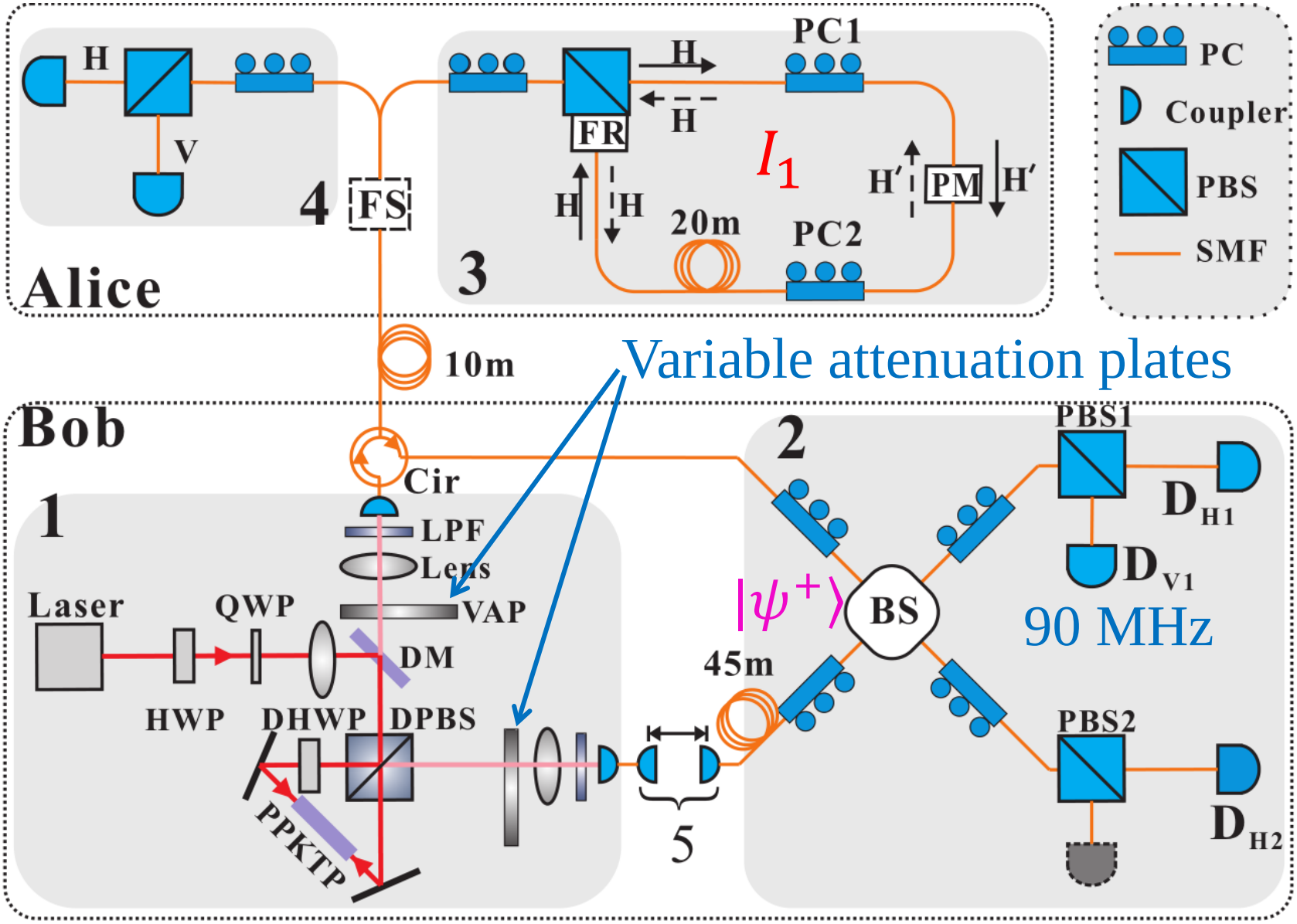
Coding operations

$\frac{V_1}{V_\pi}\pi$	$\frac{V_2}{V_\pi}\pi$	**Operations**	***e*** **Error rate in 1h.(avg.)**
0	0	I_0	5.4%
π	π	I_1	6%
0	π	Y_0	4.1%
π	0	Y_1	4%

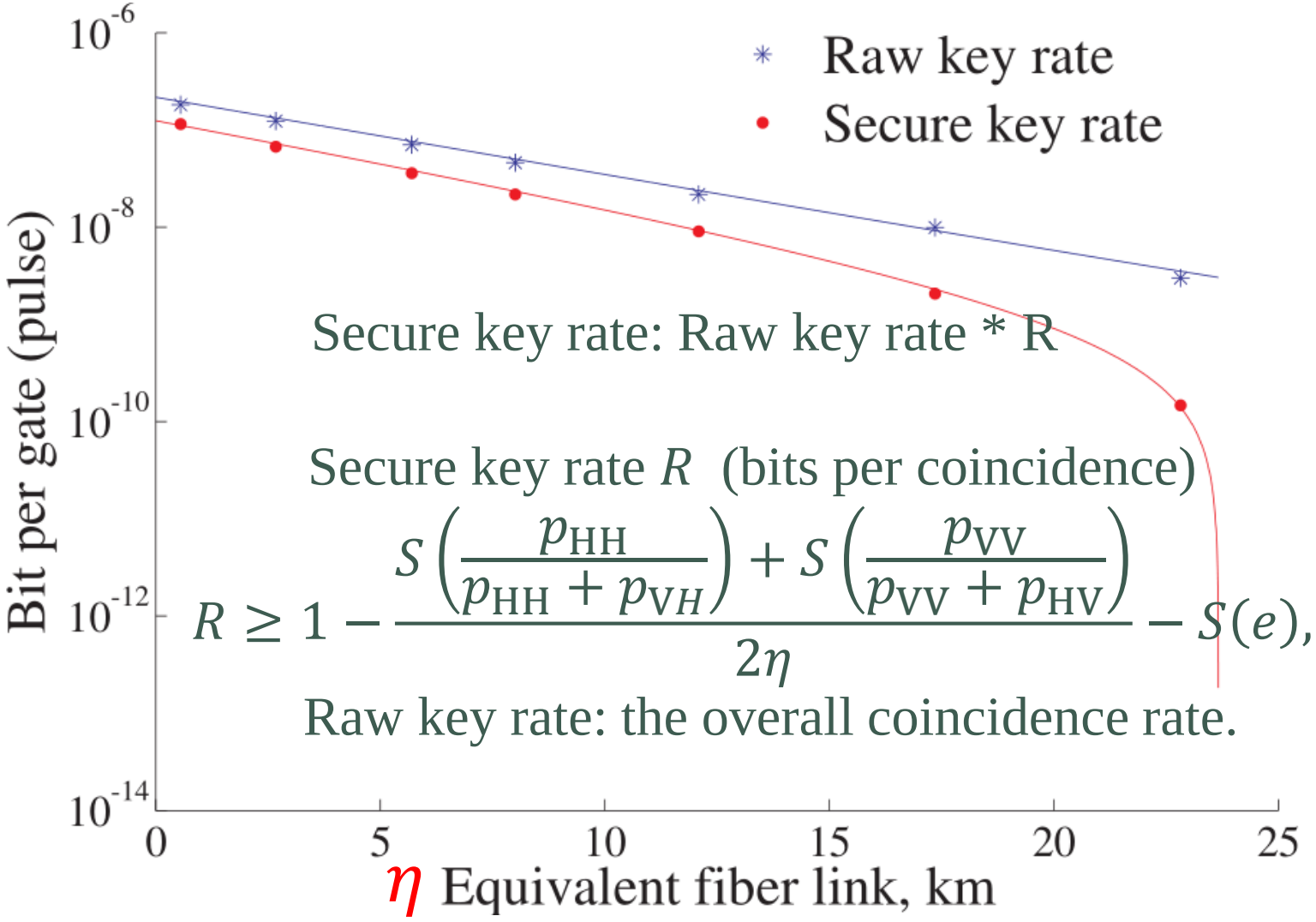
$|\psi^-\rangle \rightarrow |H\rangle_h|V\rangle_t - e^{i\frac{V_1-V_2}{V_\pi}\pi}|V\rangle_h|H\rangle_t$ at BSM



THE LOSS-TOLERANT TEST



Secure key rate



Secure key rate: Raw key rate * R

Secure key rate R (bits per coincidence)

$$R \geq 1 - \frac{S\left(\frac{p_{\mathrm{HH}}}{p_{\mathrm{HH}} + p_{\mathrm{VH}}}\right) + S\left(\frac{p_{\mathrm{VV}}}{p_{\mathrm{VV}} + p_{\mathrm{HV}}}\right)}{2\eta} - S(e),$$

Raw key rate: the overall coincidence rate.

Conclusion

1. Verify the feasibility of modified Ping-Pong protocol over lossy telecom-fiber channels.

I. The first demonstration of secure entanglement-based deterministic QKD over lossy channels.

II. The first demonstration of Ping-Pong protocol on telecom fiber.

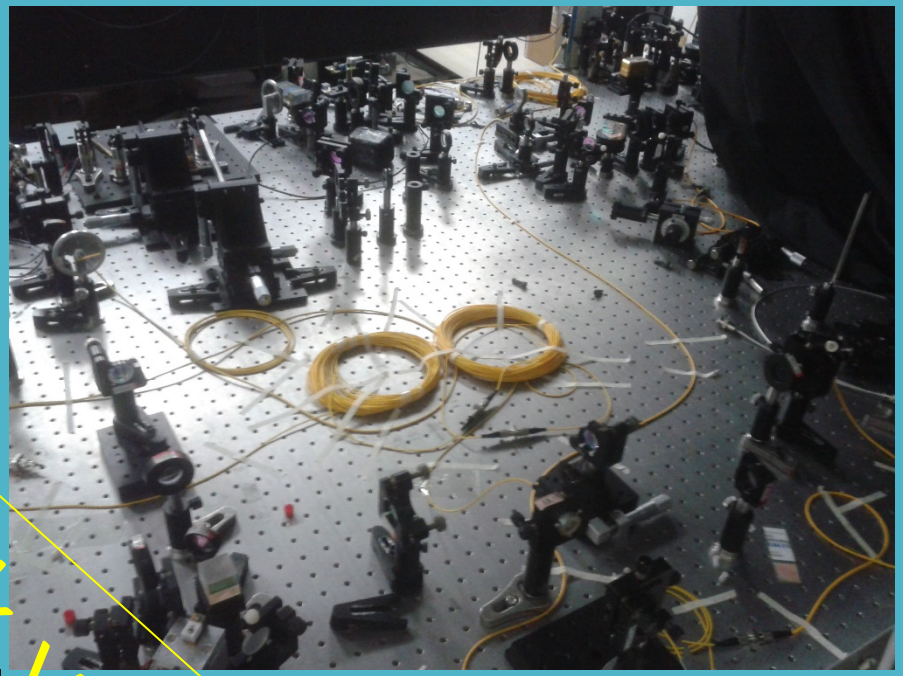
2. Build a high-quality entangled photon source at degenerate wave of 1560 nm using PPKTP-Sagnac configuration.

3. Design a polarization Sagnac interferometer for coding.

I. Merits: low loss, stability and compactness

II. Allow that the backward and forward transmission channels share the same fiber link, hence reduce the difficulty and cost.

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Thank you

